

Wall-Following Autonomous Robot

Project Overview:

This project requires students to design and implement an autonomous mobile robot that can:

- Follow walls inside a predefined track without leaving it
- Handle sharp 90° turns only
- Detect each turn and classify it as Left or Right
- Count the total number of turns
- Send the turn data to a PC using WiFi or Bluetooth
- Complete the track in the shortest possible time, accounting for any penalties

Team Composition:

- Each team shall consist of **5 members**.

Core Learning Objectives:

- Learn low-level embedded C programming **without relying on high-level Arduino libraries**
- Understand sensor interfacing and placement for wall detection (IR or ultrasonic sensors)
- Learn how to design and implement a Finite State Machine (FSM) for real-time control
- Develop skills in real-time control system design
- Understand communication protocols like UART over WiFi or Bluetooth
- Practice integrating mechanical, electrical, and software components into a functional system

System Requirements:

Functional Requirements:

- Robot must start from rest and autonomously follow walls
- Detect and handle 90° turns accurately

- Count all turns and classifies them as Left or Right
- Transmit the data to a PC in a structured format:

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Turns: 5

Sequence: L, R, R, L, L

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Non-Functional Requirements:

- **Must be programmed using register-level and drivers**
- **No high-level Arduino APIs (digitalWrite, analogRead, etc.)**
- **Real-time response is required; blocking delays must be avoided as much as possible**

Phase 1: Design & Validation (Deadline: Week 12)

This phase focuses on planning and documenting the robot design before building the actual hardware.

Robot Design Constraints:

- Maximum Width: 20 cm
- Maximum Length: 20 cm
- Wheel configuration should ensure stability and maneuverability
- Sensor placement should maximize wall detection accuracy

Track Design:

- Standard corridor width: 45 cm
- Wall height: at least 10 cm
- Turns: 90° only, no curves

Finite State Machine (FSM):

Students must create an FSM that describes robot behavior. Required components:

- FSM must include a state diagram and a transition table explaining what sensor inputs trigger state changes
- FSM should handle situations like losing the wall or detecting open spaces

Phase 1 Deliverables:

- Mechanical Design: robot dimensions, wheel placement, and sensor layout
- Electrical Design: block diagram showing connections and components
- FSM Design: state diagram, transition table, and explanation of transitions
- Algorithm Description: step-by-step explanation of wall following and turn detection
- Communication Plan: WIFI or Bluetooth selection, data format for PC reporting

Phase 1 Evaluation:

Criteria	Weight
Design feasibility	25%
FSM correctness	35%
Documentation clarity	20%
Engineering reasoning	20%

Teams must pass Phase 1 to continue to Phase 2.

Phase 2: Final Competition (Deadline: Week 14)

Rules:

- Each team has 2 trials; the best trial is counted
- Robot must autonomously complete the track and send turn data

Scoring:

- Final Score = Time + Penalties (lower is better)

Penalties:

Violation	Penalty
Collision with wall	+2 sec
Losing track for >2 sec	+5 sec

Violation	Penalty
Wrong turn detection	+3 sec
Missing turn count	+3 sec
Manual intervention	Disqualification
Failure to send data	+5 sec

Ranking Criteria:

1. Lowest final score
2. If tie → fastest raw time
3. If tie → fewer penalties

Suggested Hardware:

- Microcontroller: Tiva C, ESP32, STM32, or AVR
- Sensors: Ultrasonic or IR for wall detection
- Motors: DC motors with encoders for speed and position feedback
- Communication: ESP32 for WiFi or Bluetooth module

Documentation Guidelines:

- Write the document like an engineering report with sections:
 - Introduction
 - System Overview
 - Mechanical Design
 - Electrical Design
 - FSM and Algorithm
 - Testing Plan
- Include diagrams where necessary: FSM diagram, track layout, sensor placement, and robot design
- Provide step-by-step explanations suitable for beginners to understand control logic, turn detection, and communication flow